Comparative Analysis of COVID-19 Wave Dynamics and Short-Term Forecasting Methods Across Countries

Abstract

COVID-19 case dynamics were analyzed across four countries (United States, Brazil, Japan, and Germany) to characterize epidemic waves and evaluate short-term forecasting methods. Using publicly available data, smoothed case rates were computed and wave structures were identified via threshold-based segmentation. Four forecasting approaches were compared using time-based train/test splits restricted to active epidemic periods: naive persistence, moving average, ARIMA, and random forest regression. Models are evaluated using RMSE, MAE, MAPE, and sMAPE, along with computational efficiency. Results show that model performance depends strongly on epidemic dynamics. Simple models perform well in stable low-incidence periods, while ARIMA better captures structured temporal patterns. These findings highlight trade-offs between model complexity, interpretability, and predictive performance in epidemiological forecasting.

Introduction

The COVID-19 pandemic has highlighted the importance of accurate and timely forecasting of infectious disease dynamics. Governments and public health agencies rely on such forecasts to inform policy decisions, allocate resources, and anticipate healthcare demand. However, forecasting epidemic trajectories remains challenging due to non-stationary behavior, changing intervention policies, and differences in reporting across countries.

This study focuses on modeling and forecasting COVID-19 case trends across four countries—Brazil, Germany, Japan, and the United States—using publicly available data from Our World in Data. These countries exhibit diverse epidemic patterns, making them suitable for evaluating model performance under varying conditions. In addition to forecasting, the analysis characterizes epidemic "waves," providing insight into the duration, intensity, and temporal structure of outbreaks.

To address the forecasting task, both classical time series methods and machine learning approaches are considered. Traditional statistical models such as autoregressive integrated moving average (ARIMA) capture temporal dependencies and are widely used in epidemiological forecasting. In contrast, machine learning models, such as random forests, offer flexibility in modeling nonlinear relationships and incorporating engineered features. Simple baseline methods, including naive and moving average forecasts, are also included to provide interpretable performance benchmarks.

The objective of this study is threefold: (1) to characterize COVID-19 wave patterns across countries, (2) to compare forecasting performance across a range of models, and (3) to evaluate trade-offs between accuracy and computational efficiency. By combining statistical and machine learning approaches, this work aims to provide a practical assessment of forecasting strategies for short-term epidemic prediction.

Related Work

Time series forecasting has long been a central tool in epidemiology. Classical approaches such as the autoregressive integrated moving average (ARIMA) model, introduced by George Box and Gwilym Jenkins, model temporal dependence through autoregressive and moving average components while accounting for non-stationarity. These models have been widely applied to infectious disease data due to their interpretability and relatively low computational cost.

In addition to ARIMA, simple baseline methods such as naive persistence models and moving averages are commonly used in forecasting tasks. Despite their simplicity, these approaches can perform competitively in stable settings and serve as important benchmarks for more complex models. Comprehensive overviews of these methods are provided in standard forecasting literature, including Hyndman and Athanasopoulos (2018).

More recently, machine learning approaches have been applied to epidemic forecasting to capture nonlinear relationships and incorporate additional covariates. The random forest model, introduced by Leo Breiman, is

an ensemble learning method that aggregates predictions from multiple decision trees to improve predictive performance. In the context of COVID-19, machine learning models have been used to integrate time series features, mobility data, and intervention indicators, often achieving strong predictive performance in short-term forecasting tasks.

Several studies have applied both statistical and machine learning methods to COVID-19 data. For example, Benvenuto et al. (2020) demonstrated the use of ARIMA models for predicting early pandemic trends, while Ribeiro et al. (2020) compared multiple forecasting approaches across different countries. These studies generally find that no single model consistently outperforms others across all settings, with performance depending heavily on the characteristics of the underlying time series.

In addition to forecasting, understanding epidemic wave structure has been an important area of research. Studies such as Cacciapaglia et al. (2020) have analyzed the timing and magnitude of successive COVID-19 waves, highlighting the role of policy interventions and behavioral changes. Such analyses provide valuable context for interpreting forecasting results and understanding model limitations.

Building on this prior work, the present study combines wave characterization with a comparative evaluation of forecasting models across multiple countries, emphasizing both predictive accuracy and computational efficiency.

Methods

Data

The analysis uses the publicly available Our World in Data (OWID) COVID-19 data set, which provides daily confirmed case counts and population data for countries worldwide. The study focuses on four countries with differing epidemic trajectories: the United States, Brazil, Japan, and Germany. Daily case counts are normalized to cases per 100,000 population to enable meaningful comparisons across countries of different sizes.

Pre-Processing

Dates were converted to a standard format and observations are sorted chronologically within each country. Invalid values, such as negative case counts, were treated as missing. Missing values were handled using linear interpolation within each country. Daily case counts were converted to cases per 100,000 population. A 7-day rolling average was computed to smooth reporting noise and weekly patterns.

The data set was restricted to active periods where the smoothed case rate exceeds a predefined threshold, which removes low-incidence periods that contain limited information for forecasting, and additionally truncated at June 2023 to avoid extended trailing periods of near-zero activity that can distort evaluation.

Wave Detection

COVID-19 waves are identified using threshold-based segmentation applied to the smoothed case series. A wave is defined as a continuous period during which the 7-day rolling average exceeds a specified threshold. This approach enables consistent identification of epidemic waves across countries. For each country, wave patterns were examined in terms of timing of peaks, duration of elevated transmission, and relative magnitude of case rates. These characteristics provide context for interpreting differences in forecasting performance.

Forecasting Models

Naive persistence, moving average, ARIMA, and random forest regression were evaluated. The naive model predicts future values using the most recent observed value and serves as a baseline for comparison. The moving average model forecasts future values based on the average of recent observations within a fixed window, smoothing short-term fluctuations and capturing local trends. The auto-regressive integrated moving average (ARIMA) model is used to capture patterns over time by combining past values and past errors, with differencing applied to handle trends in the data. A simple ARIMA(1,1,1) setup is used to keep the

model easy to interpret while still capturing key dynamics. A random forest model is used as the machine learning approach. It relies on features built from the time series, including past values (such as the previous 1, 2, 3, 7, and 14 days), rolling averages and variability, simple trend measures, calendar information (like day of week and month), and vaccination data. Forecasts are generated step by step, where each prediction is fed back into the model to produce the next one.

Experimental Design

For each country, forecasting is conducted using a time-based train/test split on only informative segments on the time series, rather than low-variance periods. The data set is restricted to active periods, with the final 30 days of this filtered data used as the test set, and remaining observations used for training.

Evaluation Metrics

Forecast accuracy is evaluated using mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and symmetric mean absolute percentage error (sMAPE). Computational efficiency is assessed by measuring model training time and prediction time.

Experiments

The dataset is first summarized to confirm coverage across countries.

Table 1. Data coverage summary by country after preprocessing

Country	Start Date	End Date	Number of Observations
Brazil	2020-05-08	2023-04-29	1087
Germany	2020-03-22	2023-04-08	1113
Japan	2021-01-01	2023-05-13	863
United States	2020-03-28	2023-05-20	1149

Table 1 shows the time range and number of observations used for each country after pre-processing.

Wave Characteristics

Wave patterns vary substantially across countries in both timing and intensity (Figure 1). The United States is dominated by a single prolonged wave with a very high peak in early 2022, followed by smaller, short-lived waves with much lower case levels. Brazil exhibits more moderate but smoother waves, including a large surge in early 2022 and several short, low-intensity waves toward the end of the study period. Japan shows relatively low activity early on, followed by a sequence of increasingly large waves, culminating in a long and high-intensity wave beginning in 2022. In contrast, Germany displays more volatile dynamics, with rapid growth and a very large peak during its final wave.

Figure 1. Detected COVID-19 waves across four countries

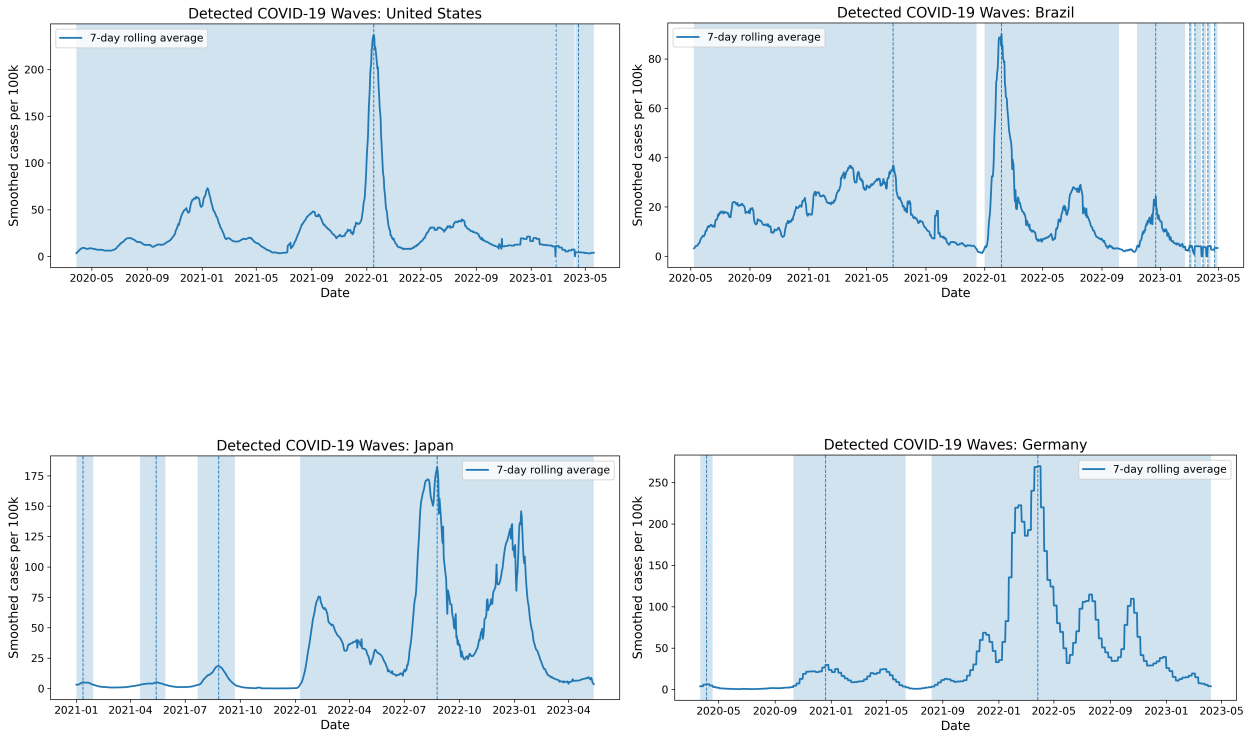


Table 2 shows clear differences in wave patterns across countries. The United States has fewer but longer-lasting waves, while Germany experiences more intense peaks with faster growth and decline. Japan falls in between, with moderate wave duration and dynamics. In contrast, Brazil has more frequent but shorter and less intense waves. These differences highlight how COVID-19 dynamics vary across countries.

Table 2. Summary of detected COVID-19 wave characteristics across countries

Country	Number of Waves	Avg Duration (Days)	Max Peak Cases	Avg Growth Rate	Avg Decline Rate
Brazil	8	126.4	90.1	0.462	0.143
Germany	3	294.0	269.9	0.560	0.324
Japan	4	156.2	182.3	0.370	0.378
United States	3	382.3	237.1	0.119	0.223

Forecasting Results

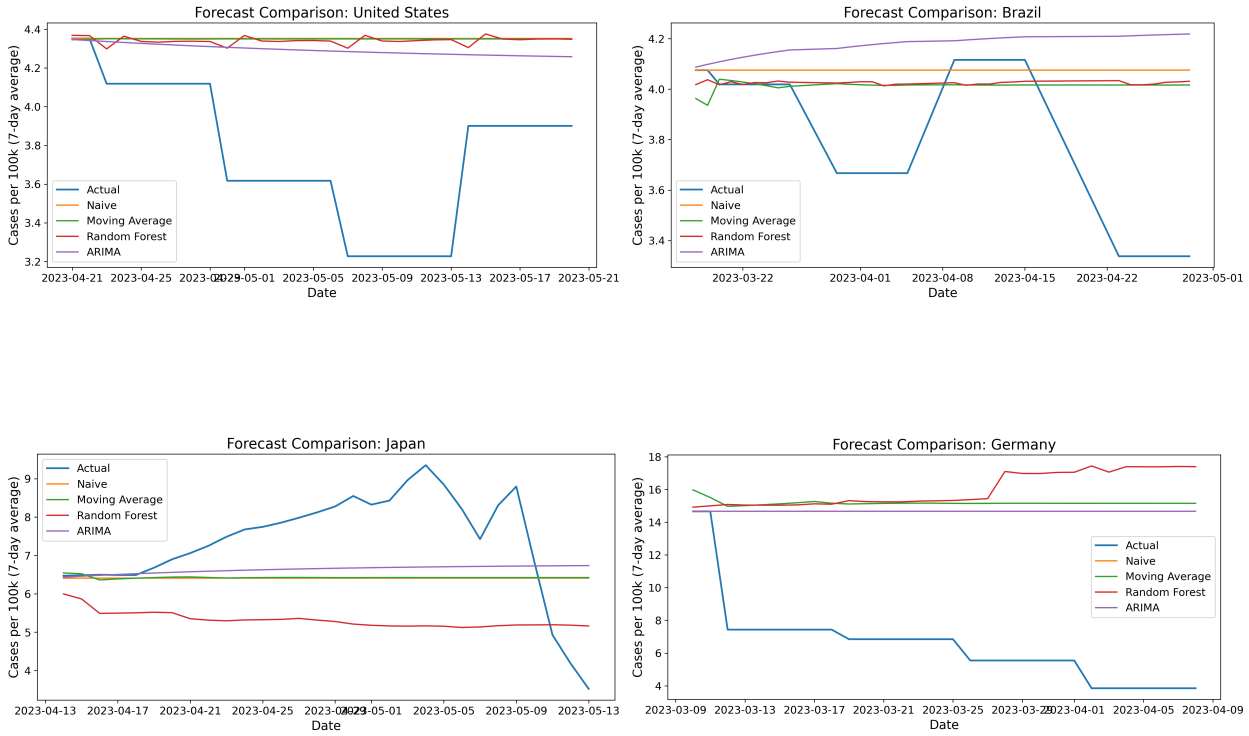
The forecasting results in Table 3 show that model performance varies by country, and no single method performs best in every setting. For Brazil, the moving average and random forest models perform similarly and achieve the lowest errors, suggesting that Brazil's case trends are relatively smooth and can be captured well by simple short-term averaging or flexible nonlinear methods. In Germany, all models perform much worse, with substantially larger RMSE, MAE, and sMAPE values, indicating that the series is harder to predict, likely because of sharper changes in case dynamics. For Japan and the United States, ARIMA performs best, with the lowest RMSE, MAE, and sMAPE among the compared models, suggesting that these countries have stronger temporal structure that is better captured by a time-series model.

Table 3. Forecasting performance across models and countries

Country	Model	RMSE	MAE	sMAPE
Brazil	Moving Average	0.274	0.373	7.272
Brazil	Random Forest	0.270	0.377	7.160
Brazil	Naive	0.290	0.408	7.658
Brazil	ARIMA	0.371	0.494	9.595
Germany	Naive	8.163	8.554	80.552
Germany	ARIMA	8.163	8.554	80.552
Germany	Moving Average	8.684	9.020	83.423
Germany	Random Forest	9.497	10.066	86.733
Japan	ARIMA	1.210	1.502	17.470
Japan	Moving Average	1.360	1.622	19.759
Japan	Naive	1.369	1.631	19.900
Japan	Random Forest	2.210	2.459	33.953
United States	ARIMA	0.536	0.636	13.743
United States	Random Forest	0.583	0.684	14.823
United States	Moving Average	0.592	0.692	15.025
United States	Naive	0.594	0.694	15.071

Figure 2 shows the forecasted versus observed case trajectories across the four countries. Overall, ARIMA produces the smoothest forecasts and generally tracks the underlying trend more closely, particularly for the United States and Japan. The moving average and naive models generate nearly flat predictions, reflecting their limited ability to adapt to short-term changes. The random forest model exhibits more variability but does not consistently capture the direction of the observed data. Across all countries, the models struggle to reproduce sharp declines in case counts, most notably in Germany and Japan, where the actual series drops rapidly while predictions remain relatively stable. This highlights a key limitation of the models in responding to abrupt changes in the time series.

Figure 2. Forecasted versus observed case trajectories across models and countries



Overall, the results suggest that simpler models are competitive when the series is stable, while ARIMA performs better when there is clearer temporal dependence. The random forest model does not consistently outperform the simpler approaches, despite being more computationally intensive.

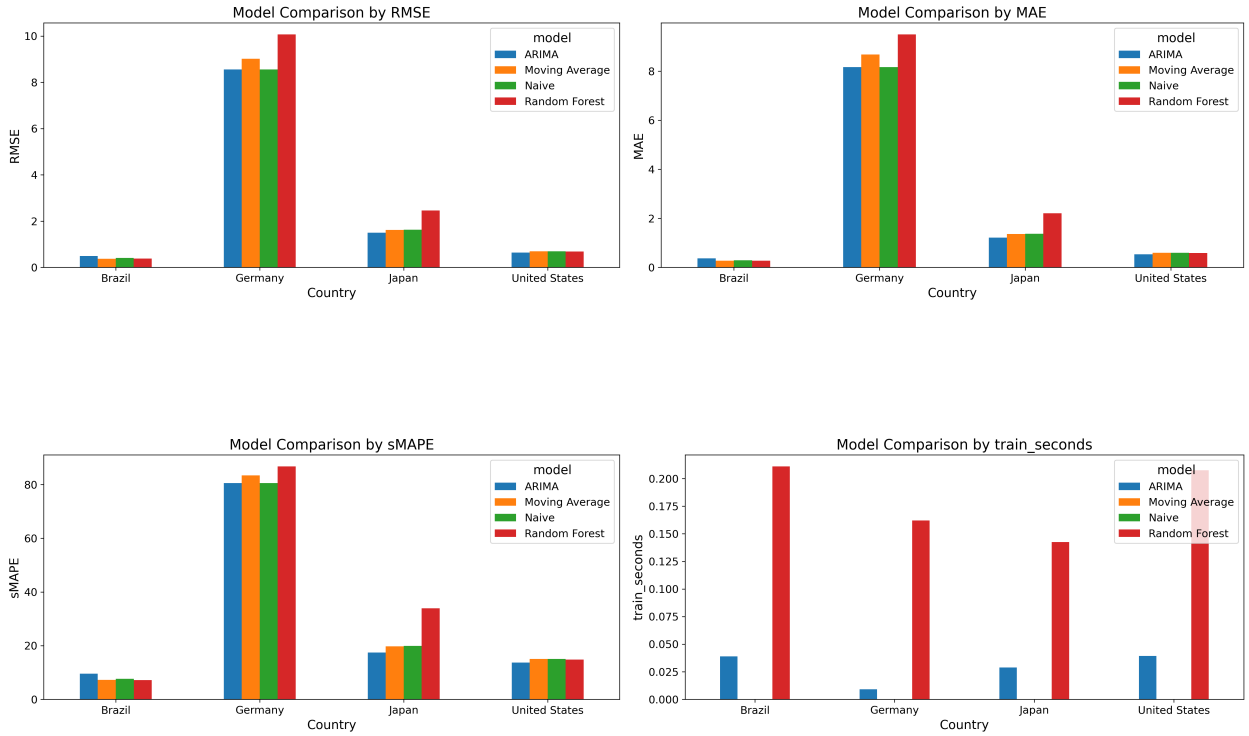
Model Comparison

Figure 3 compares forecasting performance across models using RMSE, MAE, sMAPE, and training time. Overall, model performance varies by country, with no single approach consistently outperforming others across all settings.

For Brazil, all models achieve relatively low error, with the moving average and random forest performing slightly better than ARIMA and the naive approach. This suggests that Brazil's smoother case trends can be captured well by simple averaging methods or flexible machine learning models. In contrast, Germany exhibits substantially higher error across all metrics, indicating that its case dynamics are more difficult to predict. In this setting, ARIMA and the naive model perform similarly and slightly better than the other approaches, while the random forest performs worst. For Japan and the United States, ARIMA consistently achieves the lowest RMSE, MAE, and sMAPE values, indicating that these time series have stronger temporal structure that benefits from explicit modeling of autocorrelation. The moving average and naive models perform comparably but slightly worse, while the random forest shows noticeably higher error, particularly for Japan.

In terms of computational cost, the random forest model requires the most training time across all countries, while the other models are essentially negligible in comparison. Despite this increased cost, the random forest does not provide consistent improvements in accuracy.

Figure 3. Comparison of forecasting performance across models and countries



Taken together, these results suggest that simpler statistical models remain competitive for short-term forecasting, especially when the data exhibit stable patterns, while ARIMA performs best when temporal dependence is strong.

Error Analysis

Forecasting errors vary across countries (Table 3, Figure 3), with Germany showing the largest errors across all metrics. This is consistent with the forecast trajectories in Figure 2, where models struggle to capture sharp declines. In contrast, Brazil has the lowest errors, indicating a smoother and more predictable trajectory, while Japan and the United States show moderate errors, with ARIMA performing best.

It is important to note that predictions lie within a relatively small range (approximately 0-18 cases per 100,000). As a result, even small absolute deviations can lead to large relative errors, particularly for sMAPE. Overall, forecasting accuracy is driven more by the volatility of the time series than by model complexity, with all models struggling during periods of rapid change.

Computational Efficiency

Computational efficiency varies across models, as summarized in Table 4 and illustrated in Figure 3. The naive and moving average models require essentially no training or prediction time, making them the most efficient approaches. ARIMA introduces a small computational cost due to parameter estimation but remains relatively fast overall. In contrast, the random forest model has substantially higher training and prediction times, reflecting the added complexity of feature engineering and ensemble learning.

Table 4. Average training and prediction time by forecasting model

Model	Avg Train Time (s)	Avg Predict Time (s)
Moving Average	0.000	0.000
Naive	0.000	0.000
ARIMA	0.029	0.001
Random Forest	0.181	0.806

Despite this increased computational cost, the random forest model does not consistently improve forecasting accuracy compared to simpler approaches. This highlights a clear trade-off between model complexity and efficiency. In practice, simpler statistical models may be preferable when computational resources or runtime are limited, while ARIMA provides a reasonable balance between performance and cost.

Conclusions

This study examined COVID-19 wave dynamics and short-term forecasting performance across four countries with distinct epidemic patterns. The results show that wave characteristics differ substantially in terms of duration, intensity, and growth dynamics, highlighting the importance of country-specific context when modeling infectious disease trends (Figure 1, Table 2).

Across forecasting models, no single method consistently outperformed others in all settings. Simpler approaches such as the naive and moving average models performed competitively in more stable periods, while ARIMA provided the most reliable performance in countries with stronger temporal structure, particularly in Japan and the United States (Table 3, Figure 3). The random forest model, despite its flexibility, did not consistently improve accuracy and required substantially greater computational resources.

Error patterns further emphasize that forecasting performance is driven more by the underlying volatility of the time series than by model complexity. All models struggled to capture rapid changes, especially sharp declines, and even small absolute errors were magnified due to the relatively low range of case values (approximately 0-18 cases per 100,000).

Overall, the findings highlight a clear trade-off between model simplicity, interpretability, and computational efficiency. For short-term epidemiological forecasting, simpler statistical models remain strong baselines, while ARIMA offers a practical balance between accuracy and cost. Future work could explore hybrid or adaptive approaches that better capture abrupt changes in epidemic dynamics and improve robustness across varying conditions.

References

Ritchie, H., Mathieu, E., Rodés-Guirao, L., et al. (2020). *Coronavirus Pandemic (COVID-19).* Our World in Data.
https://ourworldindata.org/coronavirus

Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time Series Analysis: Forecasting and Control.* Wiley.

Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and Practice.* OTexts.
https://otexts.com/fpp3/

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32.

Benvenuto, D., Giovanetti, M., Vassallo, L., Angeletti, S., & Ciccozzi, M. (2020). Application of the ARIMA model on the COVID-2019 epidemic dataset. *Data in Brief*, 29.

Ribeiro, M. H. D. M., da Silva, R. G., Mariani, V. C., & dos Santos Coelho, L. (2020). Short-term forecasting COVID-19 cumulative confirmed cases: Perspectives for Brazil. *Chaos, Solitons & Fractals*, 135.

Cacciapaglia, G., Cot, C., & Sannino, F. (2020). Second wave COVID-19 pandemics in Europe: a temporal playbook. *Scientific Reports*, 10.